Deploy a Custom Apache Web Server on Amazon EC2

Simple step-by-step guide (Amazon Linux 2023 + Apache + User Data)

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1. What you will do

Launch one EC2 instance that installs Apache (httpd) and publishes a custom web page **automatically** when it starts, using a User Data script. Then open the instance's public IP in a browser to see the page.

Item	Value
Instance name	Custom-Apache-Web-Server
AMI	Amazon Linux 2023
Instance type	t2.micro (use t3.micro if t2.micro is not offered for your account)
Key pair	Proceed without a key pair
Security group	HTTP 80 from Anywhere (0.0.0.0/0 and ::/0); SSH 22 from My IP
Public IP	Auto-assign enabled
Time needed	About 10 minutes

2. Step 1 - Launch the EC2 instance

1. Sign in to the **AWS Management Console** and select your preferred Region (top-right); stay in it for the whole project.
2. Search for **EC2** and open the **EC2 Dashboard**.
3. Click **Launch instance**.
4. **Name:** enter `Custom-Apache-Web-Server`.
5. **Application and OS Images:** choose **Amazon Linux** and select **Amazon Linux 2023 AMI** (Free tier eligible).
6. **Instance type:** choose **t2.micro** (Free tier eligible).
7. **Key pair (login):** choose **Proceed without a key pair (Not recommended)**.

Network settings (click Edit)

8. **Auto-assign public IP:** make sure it is set to **Enable**.
9. **Firewall (security groups):** select **Create security group** and name it `apache-web-sg`.
10. Rule 1 - Type **SSH**, Source type **My IP**.
11. Click **Add security group rule**: Type **HTTP**, Source type **Anywhere** (this gives 0.0.0.0/0 and ::/0 together).

Note: since no key pair is used, you will not be able to log in with SSH. That is fine for this project; the User Data script configures everything automatically.

3. Step 2 - Add the User Data script

1. On the same launch screen scroll down and expand **Advanced details**.
2. Scroll to the bottom and find the **User data** box.
3. Paste the **entire script** below (starting with `#!/bin/bash`). Do not add blank lines before it.
4. Click **Launch instance** in the right-hand Summary panel.
5. Click **View all instances** and wait until **Instance state = Running**.

```
#!/bin/bash
# Update package list and install Apache
dnf update -y
dnf install -y httpd

# Start and enable Apache service
systemctl start httpd
systemctl enable httpd

# Set permissions for Apache root directory
chown -R ec2-user:ec2-user /var/www/html

# Create a custom index.html page
cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Welcome to My Custom Apache Server</title>
    <style>
        body { font-family: Arial, sans-serif; background-color: #f4f4f4;
               text-align: center; padding: 50px; }
        .container { background: white; padding: 20px; border-radius: 10px;
                     box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.1);
                     display: inline-block; }
        h1 { color: #333; }
        p { color: #666; }
    </style>
</head>
<body>
    <div class="container">
        <h1>Welcome to My Custom Apache Web Server!</h1>
        <p>Hosted on an Amazon Linux 2023 EC2 Instance.</p>
        <p>This page was deployed automatically using AWS User Data.</p>
    </div>
</body>
</html>
EOF

# Restart Apache to apply changes
systemctl restart httpd
```

Tip: the CSS is shown on fewer lines here to fit the page. Either form works; you can also copy the original script straight from your project file.

4. Step 3 - Verify the web server

1. Wait about **2-3 minutes** after the instance is Running (the script is installing updates and Apache) and until **Status check** shows **2/2 checks passed**.
2. In **EC2 > Instances**, click the instance and copy the **Public IPv4 address** from the Details tab.

3. Open a new browser tab and go to `http://<your-public-ip>`. Use **http**, not https.
4. You should see **Welcome to My Custom Apache Web Server!** Take a screenshot for your project submission.

5. If the page does not load

Problem	What to check
Page times out	Security group has inbound HTTP 80 from 0.0.0.0/0; instance has a public IPv4 address; you typed http:// not https://.
Connection refused	Wait a few more minutes; User Data may still be running. Check **Actions > Monitor and troubleshoot > Get system log** for errors.
Default page, not custom page	The script was pasted incompletely. Relaunch the instance with the full script.
No public IP	Relaunch with Auto-assign public IP = Enable in Network settings.
My IP changed (SSH)	Only matters for SSH. Edit the SSH rule and select My IP again.

6. Clean up (avoid charges)

1. EC2 > Instances: select **Custom-Apache-Web-Server**.
2. Click **Instance state** > **Terminate (delete) instance** and confirm.
3. Optional: delete the security group `apache-web-sg` from EC2 > Security Groups once the instance shows Terminated.

7. What this project teaches

- Launching and configuring an EC2 instance from the console.
- Security groups: HTTP open to the world, SSH limited to your own IP.
- Automating setup with User Data (runs once, as root, at first boot).
- Installing and managing a service with dnf and systemctl on Amazon Linux 2023.